

Lab 16 - Designing Your Own URLs (Routing)

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Assignment: Lab 16 | ITI Summer Training - Web Development Using .NET

Fingerprinted values for Lab ID 31		
Variable Name	Formula / Derivation	Result Value
MAX_YEAR	$(\text{Lab ID mod } 4) + 1 = (31 \text{ mod } 4) + 1 = 3 + 1$	4
MIN_GPA	$2.5 + (\text{Lab ID mod } 3) * 0.5 = 2.5 + 1 * 0.5$	3.0
INTAKE_CODE	"iti" + A/B/C depending on Lab ID mod 3 (31 mod 3 = 1)	itiB

Part A - Verify and Orient

Verified that conventional routes inherited from yesterday behave properly. Added fingerprinted values header comment to Program.cs.

- **Why does the default route sit at the bottom of the table?** Specific routes must be placed above general ones because routing walks the list top-to-bottom and stops at the first match. The default catch-all route (`{controller}/{action}/{id?}`) is extremely generic and would capture specific routes like `/students/3` or `/students/top/3` if it were placed at the top.

Part B - A Second Address for a Page that Already Exists

Added rosterconventional aliasroute mapping theliteral segment**roster** to the Students.Index action. Placed it above the default route.

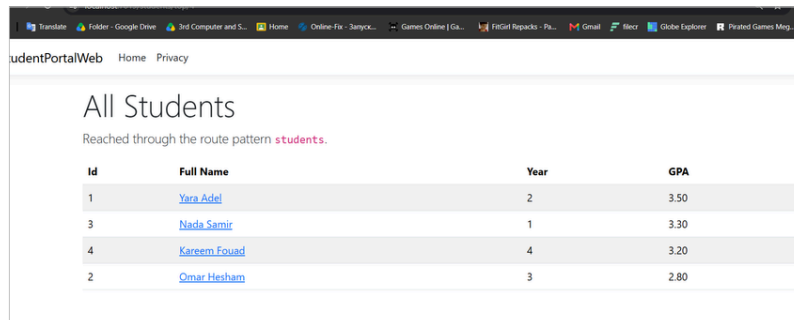
- **Is it acceptable for two different URLs to reach the same action?** Yes, aliases help with marketing or backward compatibility, provided a canonical link is defined in the HTML head to prevent duplicate content indexing penalties.

Part C - A Route with a Personalised Constraint (MAX_YEAR = 4)

Added action **Top(intcount)** to **StudentsController**, loading **studentsordered** by **GPA** descending, taking the first count of them, and passing them to **Index.html**.

Registered route **students/top/{count:int:range(1,4)}** in **Program.cs**. Added a protective guard clause to short-circuit invalid fallbacks.

- **Is your MAX_YEAR itself accepted, or rejected?** Accepted. The **range(1,4)** constraint is inclusive of both bounds, so 4 is accepted, while 5 is rejected.

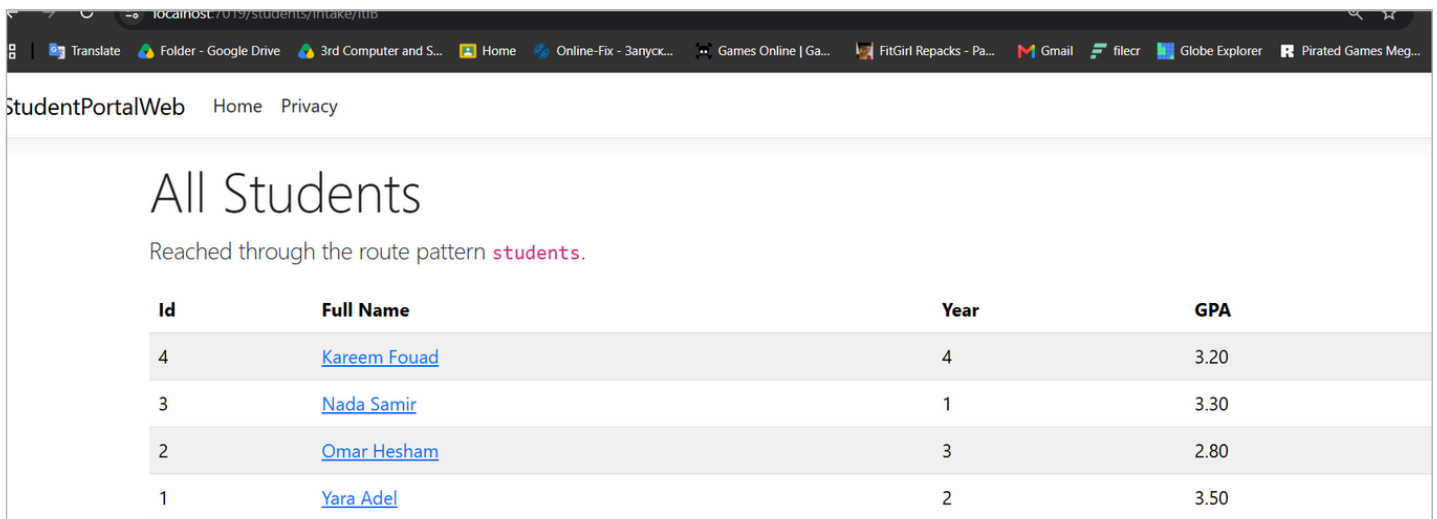


Id	Full Name	Year	GPA
1	Yara Adel	2	3.50
3	Nada Samir	1	3.30
4	Kareem Fouad	4	3.20
2	Omar Hesham	3	2.80

Part D - A Constraint You Write Yourself

Created **IntakeCodeConstraint** implementing **IRouteConstraint**. Restricts routes to 'itiB' case-insensitively, utilizing a **TryGetValue** guard clause. Registered constraint mapping as 'intakecode'.

- **Why must the constraint NOT touch the database?** Constraints run on every candidate route of every request (including static files, icons, etc.) during URL matching. Querying the database inside **Match()** would cause huge performance penalties.

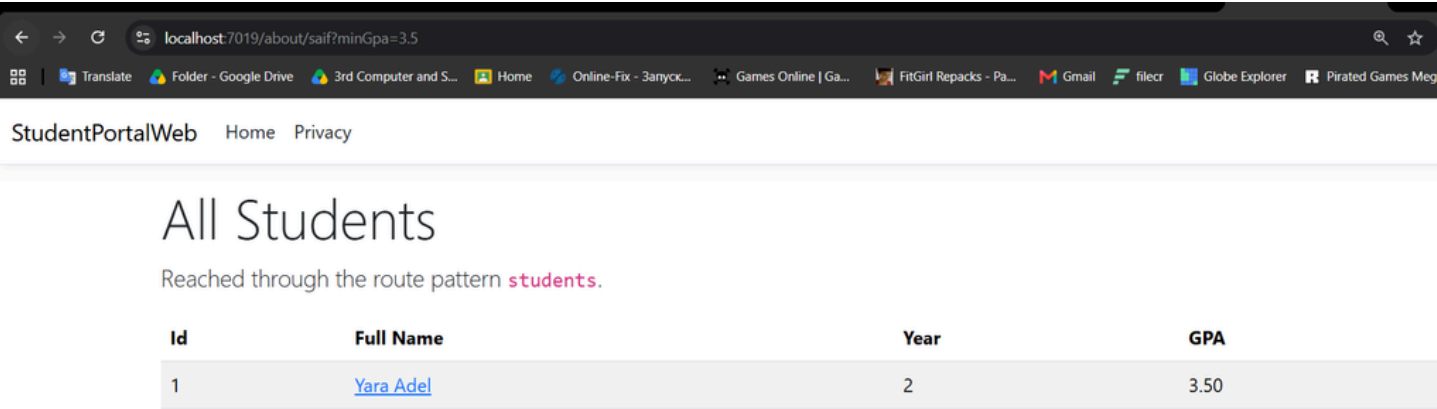
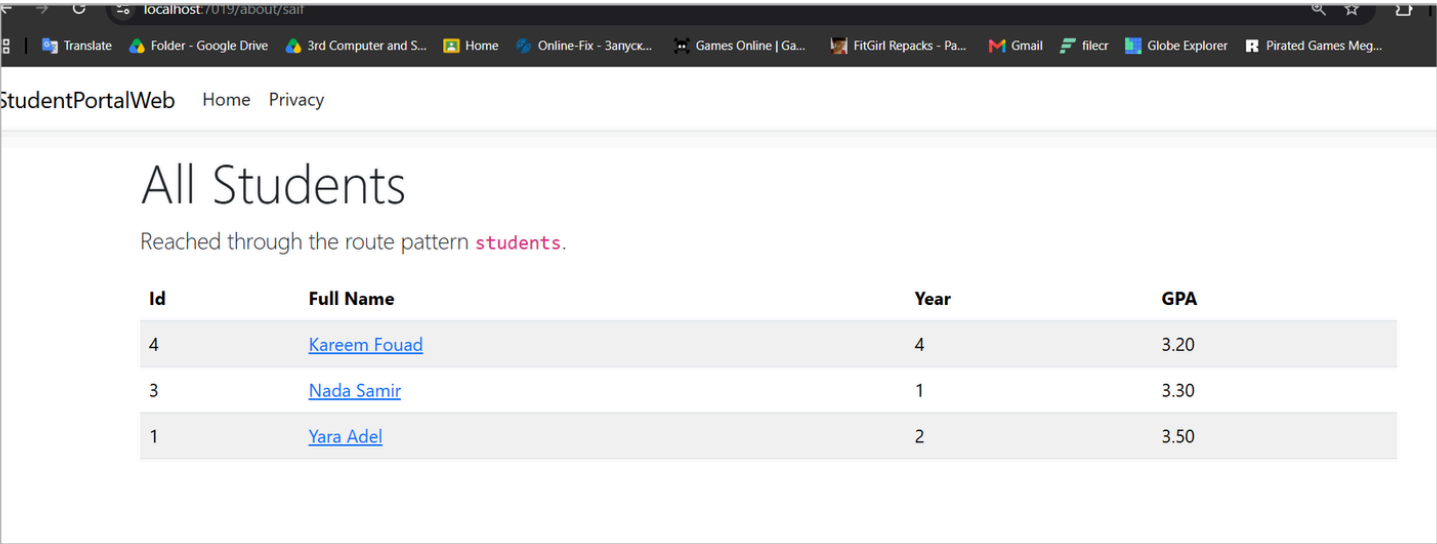


Id	Full Name	Year	GPA
4	Kareem Fouad	4	3.20
3	Nada Samir	1	3.30
2	Omar Hesham	3	2.80
1	Yara Adel	2	3.50

Part E - Your Own Address

Added action **About**([FromQuery] double? minGpa) with attribute route **[Route("about/saif")]** to StudentsController. Filters students whose GPA is >= minGpa, defaulting to 3.0 (MIN_GPA) if null.

- **Why does /Students/About return a 404?** Once an action has any attribute route, it is decoupled from conventional routing and no longer responds to conventional patterns.
- **Why does minGpa belong in the query string rather than in the path?** Because minGpa is a filtering parameter (a view of the resource) rather than an identity of a resource. /about/saif?minGpa=3.5 and /about/saif?minGpa=3.0 both return different views of the same page.



Part F - Wrap-Up Reflection

1. **Fingerprinted Values:** LabID:31 | MAX_YEAR: 4 | MIN_GPA: 3.0 | INTAKE_CODE: itiB.

2. **Request flow for /students/top/5:**

- Request enters the pipeline, custom logging middleware logs [START] Request path : /students/top/5.
- Routing middleware evaluates students/top/{count:int:range(1,4)}. Since 5 > 4, the range constraint fails, and routing skips this route.
 - It falls through to the default route {controller}/{action}/{id?}, mapping to controller=students, action=top, id=5.
 - MVC invokes Top(int count). Because parameter name count does not match id segment, count binds to its default 0.
 - Top action starts running and hits the protective guard clause count < 1 || count > 4. Since 0 < 1, it returns NotFound() (404) immediately.
- The pipeline ends, logging [END] Request path : /students/top/5. The user gets a 404.
- **What does NOT happen:** No EF database query is executed, since the request is short-circuited at the controller guard clause.

3. **Built-in 'int' vs. Custom 'intakecode':**

- **Same:** Both implement IRouteConstraint and are registered in the ConstraintMap. The routing engine calls Match() speculatively on both to accept/reject segments.
- **Different:** The 'int' constraint is pre-compiled and registered by Microsoft. The 'intakecode' constraint class is custom-written, case-insensitively checks for 'itiB', and must be registered manually in Program.cs.

4. **Attribute Route Guarantee:** It is a guarantee because it ensures strict URL contract enforcement, protecting the action from being accidentally exposed under old conventional paths like /Students/About.